BA08 « Data Science for the Smart City » – The project

Gilles MOREL – UTC – GU - March 2024

You have to develop a machine learning case study in Python with real data coming from a city or a building. You have the choice between a project level 1 and a project level 2 (see below). Project can be done preferably by teams of two students or possibly alone.

Examples of similar projects and the Jupyter tool (notebook) will be presented in the chapter on Python tools. The results of the projects will be presented in the last session of the course. The writen report is the Jupyter notebook itself.

Project level 1 : development of a smart city / machine learning use case « from scratch »

From real data produced by a city or a building (see annex 1), you have to imagine and specify a use case using a machine learning approach and develop the « proof of concept » with the Python tools, including the main steps of a classical machine learning study presented in the course.

For the data and the use case, you have the choice between :

- a use case and data provided by an UTC partner (still in discussion and preparation)
- data coming from an external site, especially open data provided by cities in the world (see a provisional list in annex 1) – Those data can be spatial (cartography), temporal (time series), or from others types and origins.

Project level 2 : analysis and improvement of an existing smart city / machine learning use case

You first have to choose an already developed use case (with Python / Jupyter) that is relevant with the course and that you can find either on DataCamp or Kaggle (or another site - A list of examples is given in annex 2.). You have to valid the project with the course coordinator.

Then, you have to analyse and understand all the steps of the study in detail, both on the level of the methodology, the technical choices and the Python code.

Then you must be able to « replay » the case in the Jupyter environment, present and explain it to an audience.

Then, from this initial basis, you must propose and justify one or several modifications and improvements of the process and code, implement the variant(s) and evaluate and compare the score with the previous study.

The modifications can lay on any phase of the study : data preprocessing, feature selection/extraction, dimensionality reduction, alternative model (and comparaison), hyper-parameter tuning … You must keep the initial process and just modify some steps and slice of code.

Finally, you must present the initial case, your proposal, and evaluate their relevance.

Annex 1 - Open data smart cities in the world

You must find the open data portail of the city, analyse the data and then imagine a service using part of the data, possibly completing with other sources (cartography, meteorology …)

France

- Paris
- Marseille
- Lyon
- Cannes
- Dijon
- Rennes
- Tours
- Toulouse
- Nantes

Europe (out of France)

- London
- Dublin
- Copenhaguen
- Luxemburg
- Oslo
- Stockholm
- Helsinki

North America - USA

- San Diego
- Chicago
- Los Angeles
- NYC
- San Francisco
- Seattle
- Jersey city

Canada

- Montreal
- Ottawa
- Toronto
- Vancouver

Asia

- Dubaï
- Singapour

Australia

- Brisbane

Annex 2 : selection of projects with Jupyter on Kaggle

Mobility – Traffic

Bike Sharing Demand | Kaggle

Traffic Prediction and some Analysis | Kaggle

Energy

Energy Consumption Forecast | Kaggle

Energy consumption in NL with GIS data | Kaggle

Water

https://www.kaggle.com/juanandressimonetti/cmp-acea-smart-water-analytics

Housing

NYC Airbnb: EDA, Visualization, Regression | Kaggle

Security

Bangalore Road Accidents : Basic Data Analysis | Kaggle

Crimes in Boston | Multiclass / Clustering | Kaggle

Smart building

Smart Building System Occupancy Prediction | Kaggle

A room has occupants or not? | Kaggle

Environment - climate

Solar Radiation Prediction | Kaggle